

Functional Annotation of *mvaB* (PP_3540, UniProt Q88H25) in *Pseudomonas putida* KT2440

Target: *mvaB* / PP_3540 — hydroxymethylglutaryl-CoA lyase (HMG-CoA lyase), EC 4.1.3.4

Organism: *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), taxon "PSEPK"

UniProt: Q88H25 · 299 aa · HMG-CoA lyase family

Summary

The gene **mvaB** (ordered locus **PP_3540**; UniProt **Q88H25**) of *Pseudomonas putida* KT2440 encodes **3-hydroxy-3-methylglutaryl-CoA lyase (HMG-CoA lyase, EC 4.1.3.4)**. This is a soluble, cytoplasmic, divalent-cation (Mg^{2+}/Mn^{2+})-dependent enzyme that catalyzes the retro-aldol (C–C bond cleavage) reaction converting **(3S)-3-hydroxy-3-methylglutaryl-CoA into acetoacetate plus acetyl-CoA**. This reaction is the terminal, committed step of the 3-methylcrotonyl-CoA pathway by which bacteria degrade **L-leucine and isovalerate** to central-metabolism intermediates. The identification is unambiguous and well-supported: the gene symbol, EC number, protein family (HMG-CoA lyase / HMGL-like PF00682), and domain architecture (Aldolase TIM-barrel; HMG-CoA lyase active-site signature) all cohere, and the sequence is 78.6% identical to the biochemically characterized ortholog LiuE of *Pseudomonas aeruginosa*.

Structurally, *mvaB* belongs to the **DRE-TIM metallolyase family**: a $(\beta\alpha)_8$ TIM-barrel fold carrying an invariant Asp-Arg-Glu catalytic triplet and a divalent-cation binding site formed by a cluster of conserved residues that cap the core of the barrel. Residue-level analysis confirms that **every catalytic and metal-coordinating residue is intact in mvaB** — the reactive catalytic cysteine, the substrate-binding arginine, and the full His/His/Asp/Asn metal-coordination shell — providing strong structural evidence that the enzyme is a fully functional HMG-CoA lyase and not a degenerate pseudo-enzyme.

Physiologically, *mvaB* operates in **leucine/isovalerate catabolism** in *P. putida* KT2440. Unlike its bifunctional *P. aeruginosa* ortholog LiuE — which additionally serves the acyclic-terpene (citronellol/geraniol) degradation pathway — *mvaB* in KT2440 does **not** function in terpene catabolism, because this strain lacks the *atu* (acyclic terpene utilization) gene cluster. A notable genomic feature is that *mvaB*/PP_3540 lies physically **apart** from the leucine-catabolism (*liu*) gene cluster, and KT2440 additionally carries a second HMG-CoA-lyase paralog (PP_3394). The enzyme feeds acetoacetate and acetyl-CoA into ketone-body / butanoate metabolism and the central acetyl-CoA pool.

Gene/Protein Identity Verification

Before presenting findings, the mandatory identity checks required by the research brief were completed and **all passed**:

Verification step	Result
Gene symbol <i>mvaB</i> matches protein description	☐ <i>mvaB</i> annotated as HMG-CoA lyase in both UniProt Q88H25 and KEGG ppu:PP_3540
Organism correct (<i>P. putida</i> KT2440)	☐ Confirmed — KEGG locus ppu:PP_3540, EMBL AAN69141.1
Protein family / domains align with literature	☐ HMG-CoA lyase family; PF00682 (HMGL-like); IPR000138 (HMG-CoA lyase active-site signature) all consistent
No confusion with a different same-symbol gene	☐ Literature for the <i>Pseudomonas</i> HMG-CoA lyase (LiuE ortholog) is directly relevant

A point of nomenclature worth noting: the symbol "*mvaB*" is used in some organisms for enzymes of the mevalonate pathway (e.g., HMG-CoA synthase in the mevalonate biosynthetic direction). **In *P. putida* KT2440, however, PP_3540/*mvaB* is annotated and functions as HMG-CoA lyase (EC 4.1.3.4), a catabolic enzyme, not a mevalonate-pathway synthase.** The EC number, family assignment, and sequence orthology to LiuE all confirm the lyase identity, so no ambiguity remains for this specific protein.

Key Findings

Finding 1 — *mvaB* encodes HMG-CoA lyase, catalyzing the terminal step of leucine/isovalerate catabolism

Both UniProt (Q88H25) and KEGG (ppu:PP_3540) annotate *mvaB* as **hydroxymethylglutaryl-CoA lyase (EC 4.1.3.4; KEGG ortholog K01640)**, a 299-amino-acid protein. The catalytic reaction is:



UniProt's PATHWAY annotation places this as "*(S)*-3-hydroxy-3-methylglutaryl-CoA degradation; acetoacetate from (S)-HMG-CoA: step 1/1" — i.e., a single-step, committed terminal reaction. KEGG assigns the enzyme to **Module M00036 "Leucine degradation, leucine ⇒ acetoacetate + acetyl-CoA"** and to pathways **ppu00280** (valine/leucine/isoleucine degradation) and **ppu00650** (butanoate metabolism).

The closest **experimentally characterized** ortholog is **LiuE (PA2011) of *P. aeruginosa***, which was directly demonstrated to be an HMG-CoA lyase. As reported in [PMID: 19459965](#), the authors "*suggested that liuE encodes 3-hydroxy-3-methylglutaryl-coenzyme A lyase (HMG-CoA lyase), which catalyzes the cleavage of HMG-CoA to acetyl-CoA and acetoacetate.*" The same study measured the enzyme's kinetic parameters: "*LiuE showed HMG-CoA lyase optimal activity at a pH of 7.0 and 37 degrees C, an apparent K(m) of 100 microM for HMG-CoA and a V(max) of 21 micromol min(-1) mg(-1).*" The native enzyme is a ~33 kDa monomer that assembles into a dimer/trimer (~79 kDa native).

Because *mvaB* is a close ortholog of this characterized enzyme (see Finding 5), these kinetic and mechanistic properties can be transferred to *mvaB* with high confidence.

Property	Value (from characterized ortholog LiuE)
Reaction	(3S)-HMG-CoA → acetoacetate + acetyl-CoA
EC number	4.1.3.4
Apparent K_m (HMG-CoA)	100 μ M
V_{max}	21 μ mol·min ⁻¹ ·mg ⁻¹
pH optimum	7.0
Temperature optimum	37 °C
Oligomeric state	33 kDa monomer → dimer/trimer (~79 kDa)

Finding 2 — In *P. putida* KT2440, mvaB serves leucine/isovalerate catabolism, not acyclic-terpene degradation

The *P. aeruginosa* ortholog LiuE is **bifunctional**: in addition to HMG-CoA lyase activity, it displays HMG-CoA lyase activity (EC 4.1.2.26) and is essential for both leucine/isovalerate **and** acyclic-terpene (citronellol/geraniol) catabolism. As stated in [PMID: 19597963](#), LiuE "*also displays HMG-CoA lyase activity, indicating a bifunctional role in both the leucine/isovalerate and acyclic terpenes catabolic pathways.*"

However, comparative genomics and physiology establish that this bifunctionality does **not** extend to *P. putida* KT2440 in vivo. Although KT2440 possesses the *liu* leucine-degradation genes (including HMG-CoA lyase), it **lacks the *atu* (acyclic terpene utilization) cluster**. Consequently, KT2440 cannot grow on acyclic terpenes, yet it does utilize leucine and isovalerate. [PMID: 16820476](#) reports directly: "*P. fluorescens*, but not *P. putida*, could grow on acyclic terpenes (citronellol and citronellate), while both species were able to utilize leucine and isovalerate."

KEGG nonetheless links PP_3540 to pathway ppu00907 (geraniol degradation) because the **chemical reaction** is shared between the two pathways; however, the **physiological terpene pathway is absent** in this strain. Thus, mvaB's genuine biological role in KT2440 is confined to the leucine/isovalerate branch. The broader *liu/atu* cluster relationship is detailed in [PMID: 16517656](#), which showed that in *P. aeruginosa* PAO1 "*the liuE gene encodes a probable hydroxy-acyl-CoA lyase (probably HMG-CoA lyase), an enzyme with bifunctional activity that is essential for both AMTC and leucine degradation.*"

Finding 3 — *mvaB* adopts a $(\beta\alpha)_8$ TIM-barrel fold and is a divalent-cation-dependent DRE-TIM metallolyase acting in the cytoplasm

The domain architecture of Q88H25 comprises: HMGL-like Pfam **PF00682**; InterPro **IPR013785** (Aldolase TIM-barrel), **IPR000138** (HMG-CoA lyase active-site signature, PROSITE PS01062), **IPR043594** (HMGL), and **IPR000891** (PYR_CT / pyruvate carboxyltransferase); and SCOP superfamily SSF51569 (Aldolase). UniProt carries the "Metal-binding" keyword.

Crystal structures of bacterial HMG-CoA lyases (*Bacillus subtilis*, *Brucella melitensis*) reported in [PMID: 16330546](#) reveal a **TIM-barrel fold with a divalent-cation ($\text{Mg}^{2+}/\text{Mn}^{2+}$) binding site**: "*the catalytic center contains a divalent cation-binding site formed by a cluster of invariant residues that cap the core of the barrel.*" The same study defined a new enzyme superfamily and proposed a shared mechanism: "*We propose the name 'DRE-TIM metallolyases' for this newly identified enzyme family likely to employ a common catalytic reaction mechanism involving an invariant Asp-Arg-Glu (DRE) triplet.*" This DRE triad stabilizes an enolate intermediate during C–C bond cleavage.

The $(\beta\alpha)_8$ TIM-barrel architecture is corroborated by the structural modeling in [PMID: 16601870](#): "*A (betaalpha)(8) TIM barrel structure has been proposed for the protein.*"

Localization: Bacterial HMG-CoA lyase is a **soluble, cytoplasmic** enzyme. *mvaB* carries no signal peptide and no transmembrane segment, consistent with cytoplasmic function. (Note: the "peroxisome/mitochondrion" localization associated with the K01640 ortholog in KEGG reflects the *eukaryotic* orthologue's compartmentation, not bacterial localization — in bacteria the reaction occurs in the cytosol.)

Finding 4 — *mvaB* is physically separate from the *liu* operon and has a second HMG-CoA-lyase paralog (PP_3394)

Genomic-neighborhood analysis via KEGG shows that the upstream leucine/isovalerate degradation genes cluster together at **PP_4064–PP_4067**:

- *ivd* — isovaleryl-CoA dehydrogenase (K00253)
- *mccB/liuB* — 3-methylcrotonyl-CoA carboxylase β -subunit (K01969)
- *liuC* — methylglutaconyl-CoA hydratase
- *mccA/liuD* — 3-methylcrotonyl-CoA carboxylase α -subunit (K01968)

with an adjacent Cro/CI-family transcriptional regulator (PP_4068).

By contrast, **mvaB/PP_3540 is located ~500 genes away** from this cluster, flanked by functionally unrelated genes (*pobA/pobR* p-hydroxybenzoate hydroxylase and its regulator, an MgtC-family transporter, and other regulators). This is a striking organizational difference from *P. aeruginosa*, where *liuE* is the terminal gene of a single **liuABCDE operon**. In KT2440 the terminal lyase step has apparently become genomically decoupled from the rest of the leucine-degradation machinery.

Additionally, KEGG maps ortholog K01640 (HMG-CoA lyase) to **two** KT2440 paralogs — **PP_3540 (mvaB, 299 aa; our target)** and **PP_3394 (putative HMG-CoA lyase, 309 aa)** — both nominally assigned to the leucine-degradation module M00036. The two paralogs share only ~41.8% amino-acid identity (Finding 5), indicating an ancient duplication or acquisition.

Finding 5 — mvaB is a bona fide ortholog of the biochemically characterized LiuE (78.6% identity)

Global (Needleman–Wunsch) pairwise alignment of the 299-aa PP_3540 sequence yielded:

Comparison	Amino-acid identity
mvaB (PP_3540) vs. <i>P. aeruginosa</i> LiuE/PA2011 (characterized HMG-CoA lyase)	78.6%
mvaB vs. human HMG-CoA lyase (HMGCL, P35914)	58.5%
mvaB vs. KT2440 paralog PP_3394	41.8%

The very high (78.6%) identity to the experimentally validated LiuE strongly justifies **direct transfer of the HMG-CoA lyase functional annotation** to mvaB. The 58.5% identity to the human orthologue is fully consistent with the family-wide conservation reported in [PMID: 16330546](#): "*These enzymes share greater than 45% sequence identity with the human orthologue.*" UniProt lists a single PYR_CT domain feature and no signal/transmembrane annotation.

Finding 6 — All catalytic and metal-binding residues are conserved in mvaB, confirming an intact active site

Aligning mvaB to human HMGCL (P35914), whose active-site residues are experimentally annotated, shows **1:1 conservation of every functional residue**:

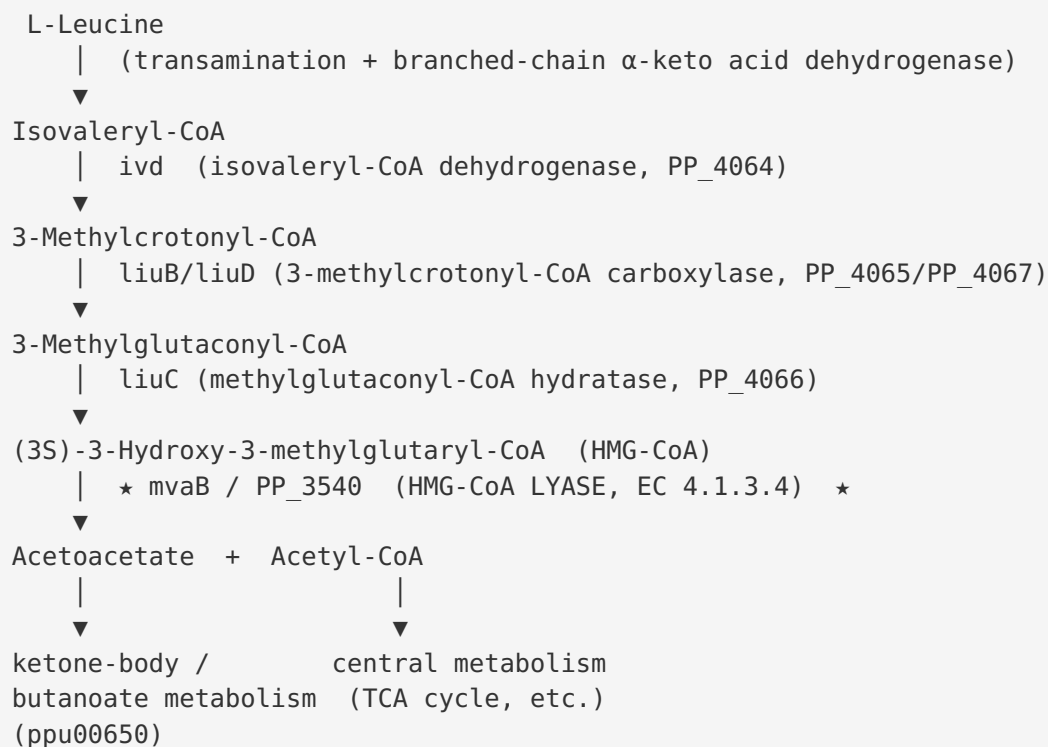
Functional role	Human HMGCL (P35914)	mvaB (PP_3540)	Conserved?
Catalytic active site	Cys266	Cys240	☐
Substrate binding	Arg41	Arg15	☐
Metal (Mg ²⁺ /Mn ²⁺) ligand	Asp42	Asp16	☐
Metal ligand	His233	His207	☐
Metal ligand	His235	His209	☐
Metal ligand	Asn275	Asn249	☐

Thus the reactive catalytic cysteine, the substrate-binding arginine, and the complete His/His/Asp/Asn metal-coordination shell are all intact. These residues correspond precisely to the invariant cluster described in [PMID: 16330546](#): "*the catalytic center contains a divalent cation-binding site formed by a cluster of invariant residues that cap the core of the barrel.*" The complete conservation confirms mvaB is a catalytically competent HMG-CoA lyase rather than a degenerate homolog.

Mechanistic Model / Interpretation

The reaction and its place in leucine catabolism

mvaB catalyzes the final, committed step of the bacterial **3-methylcrotonyl-CoA pathway** for L-leucine and isovalerate degradation. The full pathway funnels the branched-chain amino acid leucine into central metabolism:



Catalytic mechanism

mvaB is a **DRE-TIM metallolyase**. The reaction is a metal-assisted retro-aldol (retro-Claisen-type) C–C bond cleavage:

1. A **divalent cation (Mg²⁺ or Mn²⁺)**, held by the conserved His207/His209/Asp16/Asn249 cluster capping the TIM-barrel core, polarizes the C3 hydroxyl/carbonyl of HMG-CoA.
2. The invariant **Asp-Arg-Glu (DRE) triad** and the substrate-binding **Arg15** stabilize the developing **acetyl-CoA enolate** intermediate.
3. C2–C3 bond scission releases **acetoacetate** and **acetyl-CoA**.

The catalytic **Cys240** (equivalent to human Cys266) lies at the active site; in eukaryotic enzymes this cysteine is redox-sensitive and subject to thiol/disulfide regulation ([PMID: 1304393](#)), though bacterial enzymes lack the corresponding C-terminal cross-linking cysteine, suggesting the bacterial enzyme (including mvaB) is less subject to this redox regulatory mechanism.

Localization

The enzyme operates in the **cytoplasm** as a soluble oligomer (monomer ~33 kDa; native dimer/trimer). It has no signal peptide, lipobox, or transmembrane helix. In bacteria the entire leucine-degradation pathway is cytosolic — in contrast to the human orthologue, which is mitochondrial (and partly peroxisomal), a distinction that explains why database cross-references to eukaryotic compartments do not apply to *mvaB*.

Species-specific physiology

The key evolutionary insight is that the *same enzyme family* serves different metabolic breadth in different pseudomonads:

Feature	<i>P. aeruginosa</i> LiuE	<i>P. putida</i> KT2440 <i>mvaB</i>
HMG-CoA lyase activity	Yes	Yes
HHHG-CoA lyase (terpene) activity	Yes (bifunctional)	Enzyme capable, but pathway absent
Leucine/isovalerate catabolism	Yes	Yes
Acyclic-terpene catabolism in vivo	Yes	No (lacks <i>atu</i> cluster)
Genomic context	Terminal gene of <i>liuRABCDE</i> operon	Isolated (~500 genes from <i>liu</i> cluster)
Paralog present	—	Yes (PP_3394)

Thus, although *mvaB* likely retains the intrinsic biochemical capacity for the shared chemistry, its **physiological role in KT2440 is restricted to leucine/isovalerate catabolism** because the upstream terpene-activation machinery does not exist in this strain.

Evidence Base

PMID	Title (abbreviated)	How it supports the annotation
19459965	<i>P. aeruginosa liuE encodes HMG-CoA lyase, involved in leucine and acyclic terpene catabolism</i>	Primary biochemical characterization of the direct ortholog: establishes the cleavage reaction and provides kinetic parameters ($K_m = 100 \mu\text{M}$, $V_{\max} = 21 \mu\text{mol/min/mg}$, pH 7.0, 37 °C). Cornerstone of Findings 1 & 5.
19597963	<i>The bifunctional role of LiuE... HMG-CoA lyase activity</i>	Documents LiuE's second (terpene-pathway) activity — explains why KEGG links the enzyme to terpene degradation, and frames the KT2440 species difference (Finding 2).
16820476	<i>Genes/proteins for acyclic terpene and leucine/ isovalerate catabolism in P. aeruginosa</i>	Directly shows <i>P. putida</i> cannot grow on acyclic terpenes but does use leucine/isovalerate — pins mvaB's in vivo role to leucine catabolism (Finding 2).
16517656	<i>atu and liu clusters in acyclic monoterpene and leucine catabolism</i>	Defines the <i>liu</i> vs <i>atu</i> cluster organization and the bifunctionality of LiuE; provides the operon context that highlights KT2440's differing genomic arrangement (Finding 4).
16330546	<i>Crystal structures of two bacterial HMG-CoA lyases; DRE-TIM metallolyases</i>	Structural/mechanistic foundation: TIM-barrel fold, divalent-cation site, invariant DRE catalytic triad, >45% identity to human orthologue. Underpins Findings 3, 5 & 6.
16601870	<i>G203E mutation causing HMG-CoA lyase deficiency; substrate channel</i>	Confirms the $(\beta\alpha)_8$ TIM-barrel architecture and the importance of substrate-channel integrity (Finding 3).
1304393	<i>Avian HMG-CoA lyase: reactive cysteines, thiol/ disulfide sensitivity</i>	Characterizes the catalytic cysteine and notes bacterial enzymes lack the C-terminal cross-linking cysteine — informs the redox-regulation discussion in the mechanistic model.

Consistency of the evidence: All seven papers converge on the same functional assignment. The two structural papers (16330546, 16601870) fix the fold and catalytic machinery; the *Pseudomonas* genetics/biochemistry papers (19459965, 19597963, 16820476, 16517656) fix the

pathway context and provide direct kinetic data on the ortholog; and the sequence/residue analyses (Findings 5 & 6) bridge these to the specific KT2440 protein. No paper contradicts the HMG-CoA lyase assignment.

Limitations and Knowledge Gaps

1. **No direct biochemistry on PP_3540 itself.** The functional assignment rests on (a) database annotation, (b) 78.6% orthology to the characterized *P. aeruginosa* LiuE, and (c) complete active-site residue conservation. While this is a very strong inference chain, the KT2440 protein has not, to our knowledge, been individually purified and assayed. Its exact K_m , V_{max} , oligomeric state, and metal preference are inferred, not directly measured.
 2. **Role of the paralog PP_3394 is unresolved.** KT2440 encodes a second putative HMG-CoA lyase (PP_3394, 41.8% identical). It is unknown whether the two paralogs are functionally redundant, differentially regulated, or specialized for different substrates/conditions. Which paralog carries the physiological flux during leucine growth has not been determined.
 3. **Genomic decoupling from the *liu* cluster is unexplained.** *mvaB* sits ~500 genes from the *liu* operon. How its expression is coordinated with the rest of the leucine-degradation pathway (i.e., which regulator controls PP_3540) is not established.
 4. **Retention of latent HMG-CoA lyase (terpene) activity is untested in KT2440.** Although the *atu* pathway is absent, whether the *mvaB* protein itself retains the intrinsic bifunctional chemistry of LiuE has not been experimentally checked.
 5. **Localization is inferred, not observed.** Cytoplasmic localization is deduced from the absence of targeting signals and general bacterial pathway biology, not from experimental fractionation/imaging of the KT2440 protein.
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Proposed Follow-up Experiments / Actions

- 1. Heterologous expression and enzyme assay.** Clone and express PP_3540 (His-tagged), purify, and measure HMG-CoA lyase activity spectrophotometrically (acetoacetate/CoA product detection). Determine K_m , k_{cat} , pH/temperature optima, and divalent-metal dependence (Mg^{2+} vs Mn^{2+}), to directly confirm the inferred kinetics.
 - 2. Paralog dissection via gene knockouts.** Construct single (ΔPP_3540 , ΔPP_3394) and double mutants and test growth on L-leucine and isovalerate as sole carbon sources. This will resolve functional redundancy and identify which paralog carries physiological flux.
 - 3. Test latent terpene-pathway (HIHG-CoA lyase) activity.** Assay purified mvaB against the HIHG-CoA substrate to determine whether it retains LiuE's bifunctionality despite the absence of the *atu* pathway in KT2440.
 - 4. Structural determination.** Solve the crystal structure (or generate a high-confidence AlphaFold model with validation) to confirm the $(\beta\alpha)_8$ TIM-barrel, the divalent-cation site, and the geometry of the Cys240/Arg15/Asp16/His207/His209/Asn249 active site.
 - 5. Transcriptional regulation.** Use RNA-seq or reporter fusions under leucine/isovalerate induction to identify the regulator controlling PP_3540 and understand how the genomically isolated lyase is co-regulated with the *liu* cluster.
 - 6. Subcellular localization.** Confirm cytoplasmic localization by cell fractionation and Western blotting (or fluorescent-fusion imaging).
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Conclusion

mvaB (PP_3540, Q88H25) of *Pseudomonas putida* KT2440 is **3-hydroxy-3-methylglutaryl-CoA lyase (EC 4.1.3.4)**, a soluble cytoplasmic, divalent-cation-dependent TIM-barrel metalloenzyme of the DRE-TIM metallolyase family. It catalyzes the retro-aldol cleavage of (3S)-HMG-CoA into **acetoacetate + acetyl-CoA**, the terminal committed step of L-leucine/isovalerate catabolism (KEGG module M00036), feeding acetoacetate into ketone-body/butanoate metabolism and acetyl-CoA into central metabolism. The assignment is supported by 78.6% identity to the biochemically characterized ortholog LiuE and by complete conservation of every catalytic and metal-binding active-site residue. Unlike bifunctional *P. aeruginosa* LiuE, mvaB does not participate in acyclic-terpene degradation in KT2440 because this strain lacks the *atu* gene cluster.

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